TCMS software Installation Qualification – Executed

INSTALLATION QUALIFICATION SCREENSHOTS FOR TCMS APPLICATION VER. 1.0

DOCUMENT NO.: LGC/QBD/0001	TITLE : COMPUTER SYSTEM VALIDATION OF TCMS

Test Objective :	To Verify Server Identification Details.			
Test Procedure :			**Expected Result :**	
Login to the operating system with admin credentials. Open System Information & Verify Server Name & Model.			**1> System Name = QBDSER09** **2> System Model = PowerEdge R450**	
Test ID : 8.1.1	**Step No : 1.1**	**Attachment No. : 1**	**Result : Pass / Fail**	**Executed By : _____** **Date & Time : _____**

Affix / insert screenshot evidence here

Test ID 8.1.1 | Step No. 1.1 | Attachment No. 1

Test Objective :	**To Verify Specification of Server Components and Software Components.**			
Test Procedure :			**Expected Result :**	
Right Click on This PC from Desktop. Select Properties and record the Processor, Installed Memory (RAM) & Operating System.			**Processor = Intel(R) Xeon(R) Silver 4309Y CPU @ 2.80GHz 2.79 GHz (2 processors)** **Installed Memory (RAM) = 64 GB and above** **Operating System = Windows Server 2022 Standard**	
Test ID : 8.1.2	**Step No : 1.1**	**Attachment No. : 2**	**Result : Pass / Fail**	**Executed By : _____** **Date & Time : _____**

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Test ID 8.1.2 | Step No. 1.1 | Attachment No. 2

Test Objective :	**To Verify Specification of Server Components and Software Components.**			
Test Procedure :			**Expected Result :**	
Right Click on This PC from Desktop. Select Properties and record the Processor, Installed Memory (RAM) & Operating System.			**Processor = Intel(R) Xeon(R) Silver 4309Y CPU @ 2.80GHz 2.79 GHz (2 processors)** **Installed Memory (RAM) = 64 GB and above** **Operating System = Windows Server 2022 Standard**	
Test ID : 8.1.2	**Step No : 1.2**	**Attachment No. : 3**	**Result : Pass / Fail**	**Executed By : _____** **Date & Time : _____**

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Test ID 8.1.2 | Step No. 1.2 | Attachment No. 3

Test Objective :	To Verify Specification of Server Components and Software Components.			
Test Procedure :			Expected Result :	
Double click on This PC & Record the Disk partition Size.			1> C:\ Capacity = 400 GB and Above 2> D:\ Capacity = 3 TB and Above 3> E:\ Capacity = 900 GB and Above	
Test ID : 8.1.2	Step No : 2.1	Attachment No. : 4	Result : Pass / Fail	Executed By : _____ Date & Time : _____

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Test ID 8.1.2 | Step No. 2.1 | Attachment No. 4

Test Objective :	To Verify Specification of Server Components and Software Components.			
Test Procedure :			Expected Result :	
Double click on This PC & Record the Disk partition Size.			1> C:\ Capacity = 400 GB and Above 2> D:\ Capacity = 3 TB and Above 3> E:\ Capacity = 900 GB and Above	
Test ID : 8.1.2	Step No : 2.2	Attachment No. : 5	Result : Pass / Fail	Executed By : _____ Date & Time : _____

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Test ID 8.1.2 | Step No. 2.2 | Attachment No. 5

Test Objective :	To Verify Specification of Server Components and Software Components.			
Test Procedure :			Expected Result :	
Double click on This PC & Record the Disk partition Size.			1> C:\ Capacity = 400 GB and Above 2> D:\ Capacity = 3 TB and Above 3> E:\ Capacity = 900 GB and Above	
Test ID : 8.1.2	Step No : 2.3	Attachment No. : 6	Result : Pass / Fail	Executed By : _____ Date & Time : _____

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Test ID 8.1.2 | Step No. 2.3 | Attachment No. 6

Test Objective :	To Verify Specification of Server Components and Software Components.			
Test Procedure :			Expected Result :	
Double click on This PC & Record the Disk partition Size.			1> C:\ Capacity = 400 GB and Above 2> D:\ Capacity = 3 TB and Above 3> E:\ Capacity = 900 GB and Above	
Test ID : 8.1.2	Step No : 2.4	Attachment No. : 7	Result : Pass / Fail	Executed By : _____ Date & Time : _____

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Test ID 8.1.2 | Step No. 2.4 | Attachment No. 7

INSTALLATION QUALIFICATION SCREENSHOTS FOR TCMS APPLICATION VER. 1.0

DOCUMENT NO.: LGC/QBD/0001	TITLE : COMPUTER SYSTEM VALIDATION OF TCMS

Test Objective :	To Verify Specification of Server Components and Software Components.			
Test Procedure :			**Expected Result :**	
Open Control Panel & Record the Installed Application.			**1> Node.js 24.14.1 (64-Bit)** **2> PostgreSQL 16 (16.6-1)** **3> Google Chrome**	
Test ID : 8.1.2	**Step No : 3.1**	**Attachment No. : 8**	**Result : Pass / Fail**	**Executed By : _____** **Date & Time : _____**

Affix / insert screenshot evidence here

Test ID 8.1.2 | Step No. 3.1 | Attachment No. 8

Test Objective :	**To verify the system network connectivity, System Date and time format.**			
Test Procedure :			**Expected Result :**	
Open Network Connection & Verify Static IP Address.			**IP Address = 172.16.0.9**	
Test ID : 8.1.3	**Step No : 1.1**	**Attachment No. : 9**	**Result : Pass / Fail**	**Executed By : _____** **Date & Time : _____**

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Test ID 8.1.3 | Step No. 1.1 | Attachment No. 9

Test Objective :	**To verify the system network connectivity, System Date and time format.**			
Test Procedure :			**Expected Result :**	
Verify date and time format in windows system.			**Date and Time format = e.g. (24-Oct-26 Time 24 Hrs.)**	
Test ID : 8.1.3	**Step No : 2.1**	**Attachment No. : 10**	**Result : Pass / Fail**	**Executed By : _____** **Date & Time : _____**

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Test ID 8.1.3 | Step No. 2.1 | Attachment No. 10

Test Objective :	**To verify the versions of Node.js runtime, PostgreSQL, MinIO and NSSM installed for TCMS.**			
Test Procedure :			**Expected Result :**	
Open Command Prompt as Administrator. Execute the command: node -v.			**Node.js version = v24.14.1**	
Test ID : 8.1.4	**Step No : 1.1**	**Attachment No. : 11**	**Result : Pass / Fail**	**Executed By : _____** **Date & Time : _____**

Affix / insert screenshot evidence here

Test ID 8.1.4 | Step No. 1.1 | Attachment No. 11

Test Objective :	To verify the versions of Node.js runtime, PostgreSQL, MinIO and NSSM installed for TCMS.			
Test Procedure :			Expected Result :	
In Command Prompt, navigate to C:\Program Files\PostgreSQL\16\bin. Execute the command: psql --version.			psql (PostgreSQL) 16.6	
Test ID : 8.1.4	Step No : 2.1	Attachment No. : 12	Result : Pass / Fail	Executed By : _____ Date & Time : _____

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Test ID 8.1.4 | Step No. 2.1 | Attachment No. 12

Test Objective :	**To verify the versions of Node.js runtime, PostgreSQL, MinIO and NSSM installed for TCMS.**			
Test Procedure :			**Expected Result :**	
In Command Prompt, navigate to the MinIO installation directory. Execute the command: minio.exe --version.			**MinIO server version (RELEASE tag) is displayed. Record the version.**	
Test ID : 8.1.4	**Step No : 3.1**	**Attachment No. : 13**	**Result : Pass / Fail**	**Executed By : _____** **Date & Time : _____**

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Test ID 8.1.4 | Step No. 3.1 | Attachment No. 13

Test Objective :	**To verify the versions of Node.js runtime, PostgreSQL, MinIO and NSSM installed for TCMS.**			
Test Procedure :			**Expected Result :**	
In Command Prompt, navigate to the NSSM installation directory. Execute the command: nssm version.			**NSSM version = 2.24 (64-bit)**	
Test ID : 8.1.4	**Step No : 4.1**	**Attachment No. : 14**	**Result : Pass / Fail**	**Executed By : _____ Date & Time : _____**

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Test ID 8.1.4 | Step No. 4.1 | Attachment No. 14

Test Objective :	**To verify TCMS components are installed as Windows services and running.**			
Test Procedure :			**Expected Result :**	
Press Win + R, type services.msc and press Enter. Locate the PostgreSQL database service.			**Service Name = postgresql-x64-16** **Status = Running, Startup Type = Automatic**	
Test ID : 8.1.5	**Step No : 1.1**	**Attachment No. : 15**	**Result : Pass / Fail**	**Executed By : _____** **Date & Time : _____**

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Test ID 8.1.5 | Step No. 1.1 | Attachment No. 15

Test Objective :	**To verify TCMS components are installed as Windows services and running.**			
Test Procedure :			**Expected Result :**	
In Services, locate the MinIO object storage service.			**Service Name = tcms-minio** **Status = Running, Startup Type = Automatic**	
Test ID : 8.1.5	**Step No : 2.1**	**Attachment No. : 16**	**Result : Pass / Fail**	**Executed By : _____** **Date & Time : _____**

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Test ID 8.1.5 | Step No. 2.1 | Attachment No. 16

Test Objective :	**To verify TCMS components are installed as Windows services and running.**			
Test Procedure :			**Expected Result :**	
In Services, locate the TCMS application service.			**Service Name = tcms** **Status = Running, Startup Type = Automatic**	
Test ID : 8.1.5	**Step No : 3.1**	**Attachment No. : 17**	**Result : Pass / Fail**	**Executed By : _____** **Date & Time : _____**

Affix / insert screenshot evidence here

Test ID 8.1.5 | Step No. 3.1 | Attachment No. 17

Test Objective :	**To verify TCMS database creation and schema migrations.**			
Test Procedure :			**Expected Result :**	
Open SQL Shell (psql) from the Start Menu and connect as database administrator. Execute the command: \l.			**Database tcms_prod is listed.**	
Test ID : 8.1.6	**Step No : 1.1**	**Attachment No. : 18**	**Result : Pass / Fail**	**Executed By : _____** **Date & Time : _____**

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Test ID 8.1.6 | Step No. 1.1 | Attachment No. 18

Test Objective :	**To verify TCMS database creation and schema migrations.**			
Test Procedure :			**Expected Result :**	
Execute: \c tcms_prod. Execute: SELECT COUNT(*) FROM "_prisma_migrations" WHERE finished_at IS NOT NULL;. Execute: SELECT migration_name FROM "_prisma_migrations" ORDER BY finished_at DESC LIMIT 1;.			**1> Count of applied migrations = 32 2> Latest migration = 20261015120000_assignment_training_type_snapshot**	
Test ID : 8.1.6	**Step No : 2.1**	**Attachment No. : 19**	**Result : Pass / Fail**	**Executed By : _____ Date & Time : _____**

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Test ID 8.1.6 | Step No. 2.1 | Attachment No. 19

Test Objective :	**To verify MinIO object storage and TCMS document bucket.**			
Test Procedure :			**Expected Result :**	
On the server, open Google Chrome and navigate to http://localhost:9001. Login with MinIO administrator credentials.			**MinIO Console login is successful.**	
Test ID : 8.1.7	**Step No : 1.1**	**Attachment No. : 20**	**Result : Pass / Fail**	**Executed By : _____** **Date & Time : _____**

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Test ID 8.1.7 | Step No. 1.1 | Attachment No. 20

Test Objective :	**To verify MinIO object storage and TCMS document bucket.**			
Test Procedure :			**Expected Result :**	
Open Object Browser / Buckets in the MinIO Console.			**Bucket tcms-documents is listed.**	
Test ID : 8.1.7	**Step No : 2.1**	**Attachment No. : 21**	**Result : Pass / Fail**	**Executed By : _____** **Date & Time : _____**

Affix / insert screenshot evidence here

Test ID 8.1.7 | Step No. 2.1 | Attachment No. 21

Test Objective :	**To verify TCMS application installation files and configuration.**			
Test Procedure :			**Expected Result :**	
Open File Explorer and navigate to the TCMS application installation directory. Record the installation path.			**Installation directory contains: .next, node_modules, prisma, public, package.json and .env**	
Test ID : 8.1.8	**Step No : 1.1**	**Attachment No. : 22**	**Result : Pass / Fail**	**Executed By : _____** **Date & Time : _____**

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Test ID 8.1.8 | Step No. 1.1 | Attachment No. 22

Test Objective :	To verify TCMS application installation files and configuration.			
Test Procedure :			Expected Result :	
Open package.json in Notepad (read only).			1> "name" = "tcms" 2> "next" = "14.2.35"	
Test ID : 8.1.8	Step No : 2.1	Attachment No. : 23	Result : Pass / Fail	Executed By : _____ Date & Time : _____

Affix / insert screenshot evidence here

Test ID 8.1.8 | Step No. 2.1 | Attachment No. 23

Test Objective :	To verify TCMS application installation files and configuration.			
Test Procedure :			Expected Result :	
Open .env in Notepad (read only). Mask all passwords / secret keys before capturing the screenshot.			1> NEXTAUTH_URL = http://172.16.0.9:3000 2> DATABASE_URL points to localhost:5432/tcms_prod 3> STORAGE_PROVIDER = minio, MINIO_BUCKET = tcms-documents	
Test ID : 8.1.8	Step No : 3.1	Attachment No. : 24	Result : Pass / Fail	Executed By : _____ Date & Time : _____

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Test ID 8.1.8 | Step No. 3.1 | Attachment No. 24

Test Objective :	**To verify TCMS application accessibility over the network.**			
Test Procedure :			**Expected Result :**	
On a client workstation, open Google Chrome. Navigate to http://172.16.0.9:3000.			**1> TCMS Login page is displayed. 2> Browser tab title = "TCMS — Training & Competency Management System"**	
Test ID : 8.1.9	**Step No : 1.1**	**Attachment No. : 25**	**Result : Pass / Fail**	**Executed By : _____ Date & Time : _____**

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Test ID 8.1.9 | Step No. 1.1 | Attachment No. 25

IQ Execution Summary

Sr. No.	Test ID	Test Title	Attachment No(s).	Result (Pass / Fail)	Discrepancy No. (if any)	Verified By (Sign & Date)
1.	8.1.1	Server Identification	1			
2.	8.1.2	Server Components and Software Components.	2 – 8			
3.	8.1.3	Server Network Connection & Date and Time Format	9 – 10			
4.	8.1.4	Runtime and Supporting Software Version Verification	11 – 14			
5.	8.1.5	Windows Services Verification	15 – 17			
6.	8.1.6	Database Installation Verification	18 – 19			
7.	8.1.7	Object Storage (MinIO) Verification	20 – 21			
8.	8.1.8	Application Installation and Configuration Verification	22 – 24			
9.	8.1.9	Application Accessibility Verification	25			

Conclusion: Based on the above executed Installation Qualification scripts, the TCMS Application Ver. 1.0 has been installed on server QBDSER09 as per pre-approved specifications: Yes [ ] / No [ ]

Remarks: _____

Approval

Prepared by:	Designation	Sign & Date
	(Logicology)	
Reviewed by:	**Designation**	**Sign & Date**
	(Logicology)	
	(QbD Research and Development Lab Pvt. Ltd.)	
Approved by:	**Designation**	**Sign & Date**
	(QbD Research and Development Lab Pvt. Ltd.)	
Signature indicates that the information above is accurate and the validation activities proposed are in compliance with regulatory and corporate requirements.		